

LOGICAL FALLACIES

WHAT IS LOGICAL FALLACY?

An error in reasoning that occurs when invalid arguments or irrelevant points are introduced without any evidence to support them.

It is a flaw in reasoning that undermines an argument's validity, often acting as a trick or illusion of thought used to persuade.

- These errors, which can be **intentional** or **unintentional**, stem from:
 - irrelevant premises,
 - faulty evidence, or
 - incorrect inferences.

LOGICAL FALLACY

- There are two types of mistakes that can occur in arguments:
 - A **factual error in the premises**
 - The **premises fail to logically support the conclusion**

The premises fail to logically support the conclusion

- A logical fallacy is usually a mistake of this type.
- violates the principles of critical thinking because the premises do not sufficiently support the conclusion

The majority of logical fallacies involve arguments

- **Premise**-A statement in an argument that sets forth evidence
- **Conclusion**-The statement in an argument that the premises are claimed to support or imply.

Factual error in the premise:

- ✓ the mistake is not one of logic
- ✓ A premise can be proven or disproven with facts
- ✓ Example: Counting 21 when there are 23 students in class

TYPES OF LOGICAL FALLACIES

• Informal logical fallacies

- occur when there is an error in the **content** of an argument (i.e., it is based on irrelevant or false premises).
- Sub-division:
 - **Relevance**: fallacies that raise an irrelevant point
 - **Ambiguity**: the meanings of words change in the course of discussion

• Formal logical fallacies

- Occur when there is an error in the logical **structure** of an argument.

Common logical fallacies

Logical fallacies are errors in reasoning that occur when invalid arguments or irrelevant points are introduced without any evidence to support them.



Red herring fallacy
A deliberate attempt to mislead and distract an audience by bringing up an unrelated issue to falsely oppose the issue at hand



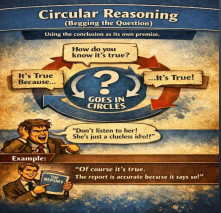
Bandwagon fallacy
Basing the validity of our argument on how many people believe or do the same thing as we do



Straw man fallacy
The distortion of an opponent's argument to make it easier to refute (e.g., by exaggerating or simplifying someone's position)



Straw Man
(Exaggerating the Argument)
"I told you I should have more respect on page 1!"
"You agree to take with all kinds!"
Example: "I really should be out there every day!"



Circular Reasoning
(Repeating the Question)
Using the conclusion as its own premise.
"How do you know it's true?"
"It's true because it's true!"
"It's true!"
Example: "Don't listen to her! She's just a foolish old woman!"
"Of course it's true. The report is accurate because it says so!"



Slippery Slope
Assuming a small step will lead to a chain of disasters.
"If we allow this, everything will fall apart!"
Example: "If we let kids redo assignments, next they'll stop working and then society will collapse!"

Common Logical Fallacies

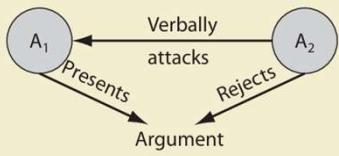
Definition:

- Attacking the person making the argument rather than the argument itself.

1. Ad Hominem (Personal Attack)

shifts focus from evidence to irrelevant personal traits.

Argument against the person



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graph LR
    A1((A1)) -- Presents --> Arg[Argument]
    A2((A2)) -- Verbally attacks --> A1
    A2 -- Rejects --> Arg
  
```

A₁ = Arguer 1
A₂ = Arguer 2
(A₂ commits the fallacy)

Example#1

❖ In his critique of Socrates in *Twilight of the Idols*, **Friedrich Nietzsche** suggests that Socrates' philosophy was merely a symptom of his physical "decadence" and being "tired of life," rather than engaging purely with the logic of Socrates' pursuit of virtue.

Example#2

❖ Television entertainer Bill Maher argues that religion is just a lot of foolish nonsense. But Maher is an arrogant, shameless, self-righteous pig. Obviously, his arguments are not worth listening to.

ad hominem abusive

➤ the second person responds to the first person's argument by verbally abusing the first person.

Example#3

❖ The Dalai Lama argues that China has no business in Tibet and that the West should do something about it. But the Dalai Lama just wants the Chinese to leave so he can return as leader. Naturally he argues this way. Therefore, we should reject his arguments.

ad hominem circumstantial

- the respondent attempts to discredit the opponent's argument by alluding to certain circumstances that affect the opponent.



Example#4

❖ Political operative Newt Gingrich has argued about the need to preserve family values. But who is he to talk? Gingrich has been married three times. He divorced his first wife while she was hospitalized for cancer, and he engaged in an extramarital affair while he was married to his second wife. Clearly, Gingrich's arguments are trash.

tu quoque ("you too")

- the second arguer attempts to make the first appear to be hypocritical or arguing in bad faith.
- Form: "How dare you argue that I should stop doing X; why, you do (or have done) X yourself."



Common Logical Fallacies

2. Straw Man

✦ distortion undermines fair debate

❑ Definition:

- ❑ Misrepresenting, exaggerating, or oversimplifying an opponent's argument to make it easier to attack.
- ❑ attack a weak version of the argument and ignore their real argument

Example:

❖ "Senator Jones wants to reduce military spending. Clearly, he wants to leave us defenseless."

The name **Straw Man** comes from the idea of a military or archery training dummy. It is much easier to knock down a man made of straw than a real person made of flesh and bone. In an argument, you "build" a straw man by creating a flimsy, distorted version of your opponent's position, knock it down easily, and then claim victory over their actual position.

Example

❖ The student status committee has presented us with an argument favoring alcohol privileges on campus. What do the students want? Is it their intention to stay boozed up from the day they enter as freshmen until the day they graduate? Do they expect us to open a bar for them? Or maybe a chain of bars all over campus? Such a proposal is ridiculous!



STRAW MAN FALLACY: EXAMPLE

Environmental Policy

Original Argument:

- ❑ "We should implement stricter regulations on industrial carbon emissions to help slow the rate of global warming."

The Straw Man:

"My opponent wants to shut down every factory in the country, destroy our economy, and force us all to live in caves without electricity. Clearly, this plan is a recipe for national disaster."

The Flaw

- The critic ignores the specific proposal (regulations) and replaces it with an extreme, absurd version (total industrial shutdown) that is much easier to attack.

Common Logical Fallacies

Definition:

- ❑ Arguing that one small step will inevitably lead to a chain of extreme consequences.
- ❑ However, no evidence is offered to prove that this chain reaction will indeed happen.

3. Slippery Slope

✚ lack of causal evidence between steps

Example:

❖ "If we allow students to redo assignments, soon they'll expect to pass without doing any work at all."

Example

- ❖ In debates over voluntary euthanasia, a common academic example of the slippery slope is the claim that allowing it for the terminally ill will "inevitably" lead to a lack of respect for human life and the eventual "extermination of the disabled".

**Slippery Slope**

Assuming a small step will lead to a chain of disasters

**Example:**

"If we let kids redo assignments, next they'll stop working and then society will collapse!"

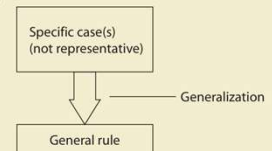
Common Logical Fallacies

Definition:

- ❑ Drawing a broad conclusion from insufficient or unrepresentative evidence.

4. Hasty Generalization

✚ importance of sample size and representativeness

Hasty generalization**Example:**

❖ "I met two rude tourists from Italy, so all Italian people must be rude."

Example

- ❖ "I traveled to a small village in the mountains of France and met three people; all three were over 80 years old and very grumpy. It's clear that the French are an aging, miserable population."

The Flaw: The sample (three people in one specific village) is far too small and localized to support a conclusion about an entire nation of over 67 million people.

Hasty Generalization

Drawing a broad conclusion from insufficient evidence



Example

- ❖ "Both Michael Jordan and LeBron James skipped or left college early and became multi-millionaires. This proves that college is a waste of time for anyone looking to be successful in business."



The Flaw:

- ❖ This identifies "outliers"—extraordinary individuals with rare talent. Generalizing their unique path to the general population (who don't have NBA-level skills) is a hasty generalization that ignores the statistical reality that a degree increases income for the average person.

Common Logical Fallacies

5. Circular Reasoning (Begging the Question)

- ✦ creates a closed loop without external justification

Definition:

- ❑ The conclusion is assumed in the premise, offering no real support.

Example:

- ❖ "Barak Obama is a good communicator because he speaks effectively."

Example

- ❖ An argument stating that "The death penalty is sinful because it is wrong and immoral" is circular. Because "wrong" and "immoral" are essentially synonyms for "sinful" in this context, the arguer is restating their claim rather than providing an external reason for it.

Circular Reasoning (Begging the Question)

Using the conclusion as its own premise.



Common Logical Fallacies

6. False Dilemma (Either/Or Fallacy)

✦ identify hidden alternatives

Definition:

- Presenting only two options when more exist.

How to spot it:

- Look for a premise that says "Either X or Y," where X and Y are extremes.
- Then ask yourself: "Is there a 'Z' that they aren't telling me about?"

Example

- ❖ "Every nation, in every region, now has a decision to make. Either you are with us, or you are with the terrorists."



The Fallacy:

- This creates a false dichotomy by erasing the possibility of **neutrality**, non-alignment, or nations that support the fight against terrorism but disagree with specific American military tactics or policies.

Common Logical Fallacies

7. Red Herring

✦ diversion weakens argumentative focus



Red herring fallacy

A deliberate attempt to mislead and distract an audience by bringing up an unrelated issue to falsely oppose the issue at hand

Definition:

- Introducing irrelevant information to distract from the main issue.

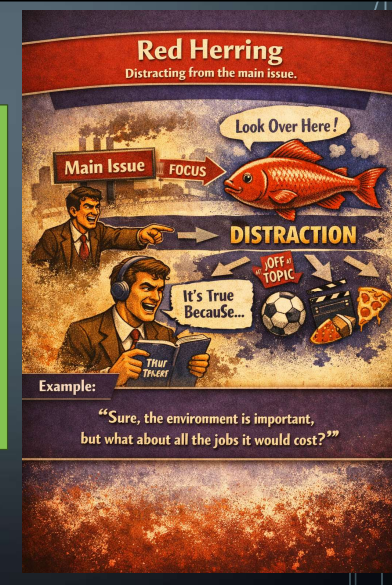
Example:

- ❖ "Why worry about pollution when there are people unemployed?"

The name **Red Herring** comes from a technique supposedly used to train hunting dogs. A dried, smoked herring (which turns red and becomes extremely smelly) would be dragged across a trail to create a powerful scent. The goal was to see if the dogs could stay focused on the original scent of the fox or if they would be distracted by the "red herring."

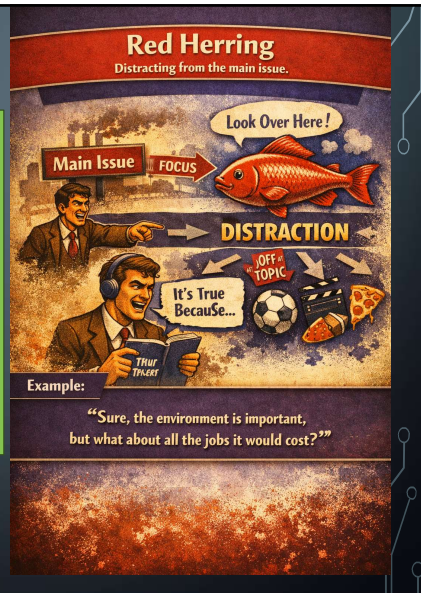
Example

- ❖ In debates regarding environmental safety, an arguer might say, "The level of mercury in seafood may be unsafe, but what will the fishers do to support their families?" This shifts the focus from public health to economic livelihood, which—while important—does not address whether the fish is safe to eat.



Example

- ❖ “In regard to my recent indictment for corruption, let’s be clear about what’s really important: unemployment! We really need to focus on creating jobs, and under my 10-point plan, here’s what we can achieve ...”



Common Logical Fallacies

Definition:

- ❑ Using an unqualified authority figure’s opinion as evidence, even when outside their expertise.

8. Appeal to Authority

- ✦ Differentiate between legitimate expertise and misplaced authority.

Example:

- ❖ “A famous actor says this diet works, so it must be effective.”

Example

- ❖ Dr. Bradshaw, our family physician, has stated that the creation of muonic atoms of deuterium and tritium hold the key to producing a sustained nuclear fusion reaction at room temperature. In view of Dr. Bradshaw’s expertise as a physician, we must conclude that this is indeed true.



Common Logical Fallacies

Definition:

- ❑ Claiming something is true or right because many people believe it.

9. Bandwagon (Ad Populum)

- ✦ Contrast popularity with validity

Example:

- ❖ “Everyone is investing in cryptocurrency, so it must be safe.”
- 1. Everyone (or most people) believes or does X.
- 2. Therefore, you should believe or do X (or X is true).



Bandwagon fallacy

Basing the validity of our argument on how many people believe or do the same thing as we do

Example

- ❖ "The vast majority of people today are switching to the iPhone. Therefore, you should switch to the iPhone too; it's obviously the best phone on the market."



Common Logical Fallacies

10. Post Hoc (False Cause)

- ✦ Reinforce the difference between correlation and causation

Definition:

- ❑ Assuming that because one event follows another, the first caused the second.

Example:

- ❖ "I wore my lucky socks and then watched the Champions League—my team won 4-0."

FALSE CAUSE VS FALSE DILEMMA

Key Differences

The main difference lies in what is being assumed by the arguer:

Feature	False Cause Fallacy	False Dilemma (False Dichotomy)
Focus	Cause and Effect.	Choices/Alternatives.
The Error	Assuming a link between two events that probably doesn't exist.	Presenting limited options as if they are the only ones.
Logic Structure	"A happened before B, so A caused B".	"Either X or Y is true; X is false, so Y is true".

Common Logical Fallacies

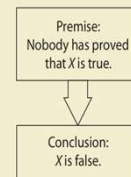
11. Appeal to Ignorance

- ✦ Lack of evidence is not evidence of absence
- ✦ Some issues are simply not yet proven or incapable of proof

Definition:

- When the premises of an argument state that nothing has been proved one way or the other about something, and the conclusion then makes a definite assertion about that thing, the argument commits an appeal to ignorance.
- The issue usually involves something that is incapable of being proved or something that has not yet been proved.

Appeal to ignorance



Example

- ❖ People have been trying for centuries to provide conclusive evidence for the claims of astrology, and no one has ever succeeded. Therefore, we must conclude that astrology is a lot of nonsense.

Appeal to Ignorance

Lack of evidence isn't proof.



Reality: "We can't conclude something is false just because it hasn't been proven true."



Common Logical Fallacies

12. Appeal to Pity

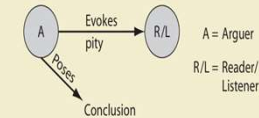
✦ Merely evoking pity doesn't make an argument logical.

✦ Emotional appeals should never replace factual or logical support for a conclusion.

Definition:

- The appeal to pity fallacy occurs when an arguer attempts to support a conclusion by merely evoking pity from the reader or listener.
- This pity may be directed toward the arguer or toward some third party.

Appeal to pity



Example

- ❖ **Taxpayer to judge:** Your Honor, I admit that I declared thirteen children as dependents on my tax return, even though I have only two. But if you find me guilty of tax evasion, my reputation will be ruined. I'll probably lose my job; my poor wife will not be able to have the operation that she desperately needs, and my kids will starve. *Surely, I am not guilty.*

Appeal to Pity

(Argumentum ad Misericordiam)

Appealing to emotions isn't proof.



Reality:

"A sad story doesn't prove guilt or innocence."

Decisions require **FACTS**, not just feelings.



LOGICAL FALLACIES

Common Errors in Reasoning

Ad Hominem

Attacking the person instead of the argument.

Example: "You can't trust him; he's a fool!"

Straw Man

Distorting an argument to make it easier to attack.

Example: "She wants to ban all cars!"

Slippery Slope

Assuming one step will lead to a disaster.

Example: "If we allow this, it will never stop!"

Hasty Generalization

Drawing a broad conclusion from too little evidence.

Example: "All teenagers are irresponsible!"

Circular Reasoning

Arguing in a circle; using the conclusion as proof.

Example: "It's true because it's true."

False Dilemma

Giving only two options when more exist.

Example: "You're either with us or against us!"

Red Herring

Distorting from the main issue.

Example: "But what about the economy?"

Appeal to Authority

Saying a claim is true because an expert says so.

Example: "A celebrity said it's the best product!"

Bandwagon

Claiming something is true because it's popular.

Example: "Everyone is doing it, so it must be right!"

Post Hoc

False Cause

Example: "I wore my lucky shirt and then aced the test!"

False Cause

Assuming that A caused B because B came after A.

Logical Fallacy



Logical Fallacy

1. What is the fallacy that occurs in this statement?

"Oh, officer, I have already had such a bad day-my husband left me, my dog was run over by the ice cream truck-it would be the end of me if you tell me that I was speeding."

Answer: Appeal to Pity

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Logical Fallacy

2. The fallacy occurs in the example is.....

"Of course you want to buy a pair of Slinky-Sox. All the fashionable people in Los Angeles and New York are wearing them. Why be out of step?"

Answer: Bandwagon [Appeal to People]

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Logical Fallacy

3. The fallacy occurs when a person exaggerates or misrepresents the opponent's argument to defeat it, as in the following:

When the National Rifle Association (NRA) claims that the arguments of those who advocate gun-control laws really intend the confiscation of all guns.

Answer: Straw Man

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Logical Fallacy

4. The fallacy occurs when the arguer diverts the attention of the reader/listener by changing the subject to some totally different issue - then draws a conclusion about this different issue and so claims to have won the argument.

Example: Students are always complaining about the food in the cafeteria. But eating well is important for academic success-study after study has shown that students who don't eat well perform poorly in their classes.

Answer: Red Herring

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Logical Fallacy

5. The fallacy in the example is most likely.....

Example: My brother Dan, the philosophy professor, says the roller coaster at the park is safe. So, it must be safe.

Answer: Appeal to Unqualified Authority

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Logical Fallacy

6. What fallacy occurs when one argues that a proposition is true simply because it has not been proved false, or false because it has not been proved true?

Answer: Appeal to Ignorance

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Logical Fallacy

7. When one, considering only unusual or atypical cases, generalizes a rule that fits them alone from particular to general as in the example below, the fallacy that occurs is.....

Example: I have met three students from SNC, and they all had blue eyes. All the students at that college must have blue eyes.

Answer: Hasty Generalization

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Logical Fallacy

- 8.** The fallacy occurs when the link between premises and conclusion depends on some imagined causal connection that probably doesn't exist
 Example: The last two times there was a Killeen lecture it snowed heavily. We'll have to cancel that lecture series if we ever want nicer weather.

Answer: False Cause

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Logical Fallacy

- 9.** The fallacy occurs when the conclusion rests on the claim that a certain event will cause a chain reaction leading in the end to some undesirable consequence, yet there is no sufficient reason to think the chain reaction will occur.

Example: The faculty are asking for a coffee machine in the lounge, but if we put in a coffee machine, they'll next want a microwave oven, then a full-sized range with a convection oven...there's no room for all these kitchen appliances in the lounge, so we'll have to turn down this request.

Answer: Slippery Slope

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Logical Fallacy

- 10.** When Galileo Galilei defended the Copernican view that the Earth revolves around the Sun, many of his detractors did not rely solely on astronomical data. His critics argued that Galileo was a "vain seeker of reputation" and a "disobedient heretic" who sought to undermine the authority of the Church.
What fallacy did the critics commit?

Answer: Ad Hominem/Attacking the Person/Argument Against the Person

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Logical Fallacy

- 11.** The fallacy in the example is
 Example: Clearly, murder is wrong, so euthanasia is morally wrong.

Answer: Circular Fallacy/Begging the Question

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Logical Fallacy

12. The fallacy in the example is

Example: "Either we cut the education budget drastically, or the state will go bankrupt".

Answer: False Dilemma
(False Dichotomy)

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Logical Fallacy

13. "A senior professor tells a grad student: 'You are free to disagree with my theory in your dissertation, but keep in mind that I am the chair of the hiring committee for the only open position in this department next year.'"

What fallacy does the Professor Commit?

A. Appeal to Pity

C. Appeal to Ignorance

B. Appeal to Emotion

D. Appeal to force

Answer: D

❖ The threat to the student's career is being used to bypass a logical debate about the theory itself.

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Thank you

